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EDUCATION

Indian Institute of Space Science and Technology
Master of Technology in Machine Learning and Computing
Department of Mathematics

Thiruvananthapuram, Kerala
Aug. 2020 – May, 2022
CGPA: 8.29/10

Jabalpur Engineering College
Bachelor of Engineering in Information Technology

Jabalpur, Madhya Pradesh
Aug. 2016 – July, 2020
CGPA: 7.44/10

EXPERIENCE

E42.AI (Light Information Systems)

Pune, India

AI Architect

Mar 2026 – Present

- **Agent Deployment Control Plane:** Built a self-serve FastAPI platform that lets teams claim, build and ship LangGraph agents without manual per-agent deployments: organisation-level agent claims, in-cluster image builds, idempotent Kubernetes deploys with auto-restore, agent validation and API-key auth.
- **Extraction CI/CD with Accuracy Gates:** Authored the CI/CD pipeline that deploys every merge request into its own namespace and evaluates document extraction on six client datasets behind a ratcheting accuracy gate, with secret and dependency scanning; introduced repo-wide linting and shipped production extraction hotfixes.
- **Enterprise RAG Chatbot:** Delivered a customer chatbot that crawls client websites alongside uploaded documents, built on an open-source RAG engine and tuned with custom LLM, embedding and reranker models, protected by guardrails.

Lead AI Engineer

Aug 2024 – Mar 2026

- **GenAI Team Leadership:** Led E42's 20-person GenAI team, mentoring engineers across the full GenAI stack, and drove agentic development across projects spanning multi-agent assistants, agent deployment infrastructure and document intelligence.
- **Document Intelligence Platform:** Architected and led the production OCR / layout / table / entity-extraction platform processing 10K+ invoices and documents daily across 10 enterprise clients at up to 98% extraction accuracy; sold as a volume-based SaaS generating up to \$80K annual revenue per client, automating invoice entry and approval follow-up with an invoice-grounded chatbot.
- **Self-Hosted LLM Serving Platform:** Led the migration (primary author) to a FastAPI serving platform with OpenAI-compatible APIs over pluggable open-source inference backends with dynamic batching and 4/8-bit quantization, plus SSE streaming, schema-constrained tool calling and managed hosted-provider fallbacks; cut end-to-end OCR+LLM latency $\sim 2\times$ (45s $\rightarrow$ ~ 22 s) by co-locating OCR and LLM on one GPU with a shared KV-cache across two model instances.
- **AR Collections Automation:** Designed an end-to-end accounts-receivable platform with AI-ranked daily collection prioritization, autonomous vendor email follow-up/reply and an analytics chatbot (email-grounded RAG and NLP2SQL with charting), plus a tone-escalating dunning engine with human-in-the-loop approval and an XGBoost ranker trained on user feedback; prototyped a headless browser agent extracting per-vendor data daily with per-invoice human verification.
- **LLM Evaluation & Observability:** Built the evaluation and observability stack: automated evaluation harnesses across 8 modules ($\sim 2,200$ eval cases), LLM tracing and instrumentation with curated datasets, an OCR evaluator (IoU + fuzzy field metrics) and retrieval-quality evals for the NLP2SQL engine.
- **LMS AI Platform:** Built the AI agents for a learning-management platform: an interactive LangGraph canvas agent, PDF/RAG search and router agents and personalised-learning suggestions, plus a vector-backed memory service carrying each learner's context across sessions.

AI/NLP Engineer

Jul 2023 – Aug 2024

- **Enterprise Multi-Agent Assistant:** Built supervisor / sub-agent architectures (CrewAI + LangGraph) behind an enterprise assistant, routing across chat, NLP2SQL and RAG; classifier, tool and RAG nodes with conversation memory, web-scraping / search / charting tools and a parallel RAG + NLP2SQL path with document grading; fine-tuned open 7B models (LoRA) for tool-calling to 83% accuracy against a GPT-4-labeled benchmark while cutting prompt overhead.
- **Invoice Extraction Pipelines:** Built the extraction framework spanning Azure Document Intelligence and open vision-language-model pipelines (top contributor to the OCR service); added VLM-based custom-entity extraction, handwriting detection, dynamic invoice/table schemas, tax-code post-processing and a field-to-bounding-box

predictor.

- **RAG Accuracy:** Raised retrieval-grounded answer accuracy from 68% by ~15 pts (semantic-match evaluation) via a reranking layer, knowledge-graph integration and a redesigned extraction + embedding-storage pipeline; largest gains on complex technical PDFs (math / physics).
- **Multimodal & Distributed Infra:** Shipped real-time streaming speech-to-text / text-to-speech (Whisper) on fault-tolerant Kafka + Kubernetes pipelines spanning OCR, RAG, speech and LLM workloads, served via hosted LLM APIs where permitted and on-prem open models for data-compliance clients, with benchmarking frameworks driving model selection.
- **Natural-Language-to-SQL Engine:** Built the NLP2SQL pipeline with a few-shot retrieval layer matching questions to curated example queries, automated schema creation and table categorisation, and chart/table rendering of results.

Circlebase (Intelligent Health Outcomes)

Bengaluru, Karnataka

NLP Engineer

Nov 2021 – Jul 2023

- **Clinical NLP Pipelines:** Developed NLP models identifying patient-related entities, social determinants of health, drug relations and adverse drug events; shipped via executable builds and integration with Django-based applications.
- **Clinical Trial Chatbot:** Built NER + relation extraction linking adverse drug events (ADEs) to drugs and matching participants with clinical coordinators by ADE severity; added a Haystack QA system with generated QA pairs, a dashboard, suggestion features and LLM integration, and dockerized it for deployment.
- **Drug Consumption Forecasting:** Performed EDA and anomaly detection on hospital data with feature selection and engineering; modeled with ARIMA, Holt-Winters exponential smoothing and XGBoost, achieving MAPE below 20% for multiple drugs.

Omdena (Internet & Jurisdiction Policy Network)

Remote

Lead ML Engineer

Sept 2021 – Nov 2021

- Built a knowledge graph with a triplet extractor algorithm that pulls info from Wikipedia articles.
- Created Datasphere using natural language processing tools like Spacy, Stanfordcore NLP, and NLTK.
- Used Datasphere to analyze and visualize big textual data, helping policymakers understand complex issues.

TECHNICAL SKILLS

- **Languages:** Python, C++, C, CUDA, SQL
- **ML / AI:** PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, vLLM, GGML, DeepSpeed
- **Infrastructure:** Docker, Kubernetes, GCP (GKE), Git, CI/CD
- **Data:** PostgreSQL, SQLite, Redis, Qdrant, Elasticsearch, Kafka

ACHIEVEMENTS

- Qualified for the ACM ICPC 2018 Kolkata-Kanpur site contest held at UIET, CSJM University, Kanpur.
- Codechef rating 1883 (amit_9); gold medal in chess, Avahan 2018 (intra-college competition).
- “So you think you can code”: organised a coding competition on Hackerearth.